

IAM & S3 Hands-On Mini Project

AWS S3 Static Website Hosting with IAM and Bucket Policy

Project Objective

The objective of this mini project was to create an IAM user with S3 permissions, create an Amazon S3 bucket, upload a static HTML file, configure public access through a bucket policy, enable S3 static website hosting, and verify that the website is accessible through the S3 website endpoint.

Technologies / AWS Services Used

- Amazon S3 — object storage and static website hosting
- AWS IAM — user and permission management
- S3 Bucket Policy — public read access for website objects
- HTML — static website content

Implementation Steps

1. IAM User Creation

An IAM user named SkillNexus-S3-User was created. The user was configured with an AWS-managed AmazonS3FullAccess policy for the hands-on exercise.

2. S3 Bucket and Static File

An S3 bucket named skillnexus-static-site-aditya was used. The static index.html file was uploaded successfully to the bucket.

3. Public Access and Bucket Policy

Block Public Access was disabled for the bucket and a bucket policy was configured to allow s3:GetObject for objects in the bucket, using a public principal.

4. Static Website Hosting

S3 Static Website Hosting was enabled with bucket hosting. AWS generated a Region-specific website endpoint for the bucket.

5. Verification

The generated S3 website endpoint was opened in a browser and the uploaded HTML page rendered successfully, confirming that the static website was accessible.

Evidence / Screenshots

The screenshot displays the AWS IAM console interface. At the top, a green banner indicates "User created successfully" with a "View user" button. The main content area shows the details for the user "SkillNexis-S3-User".

Summary:

ARN arn:aws:iam::787308165679:user/SkillNexis-S3-User	Console access Disabled	Access key 1 Create access key
Created October 01, 2026, 16:46 (UTC+05:30)	Last console sign-in -	

Permissions policies (1)

Permissions are defined by policies attached to the user directly or through groups.

Policy name	Type	Attached via
AmazonS3FullAccess	AWS managed	Directly

The interface includes a left-hand navigation menu with sections like "Access Management", "Access reports", and "Multi-account access". The bottom of the screen shows a Windows taskbar with various application icons and system status information.

1. IAM User Created

Upload succeeded

For more information, see the Files and folders table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination

s3://skillnexis-static-site-aditya

Succeeded

1 file, 211.0 B (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 211.0 B)

Find by name

Name	Folder	Type	Size	Status	Error
index.html	-	text/html	211.0 B	Succeeded	-

CloudShellAgent Toolkit for AWSFeedbackConsole Mobile App

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32°C Sunny

WindowsTaskbarIcons

ENG IN

16:54

01-10-2026

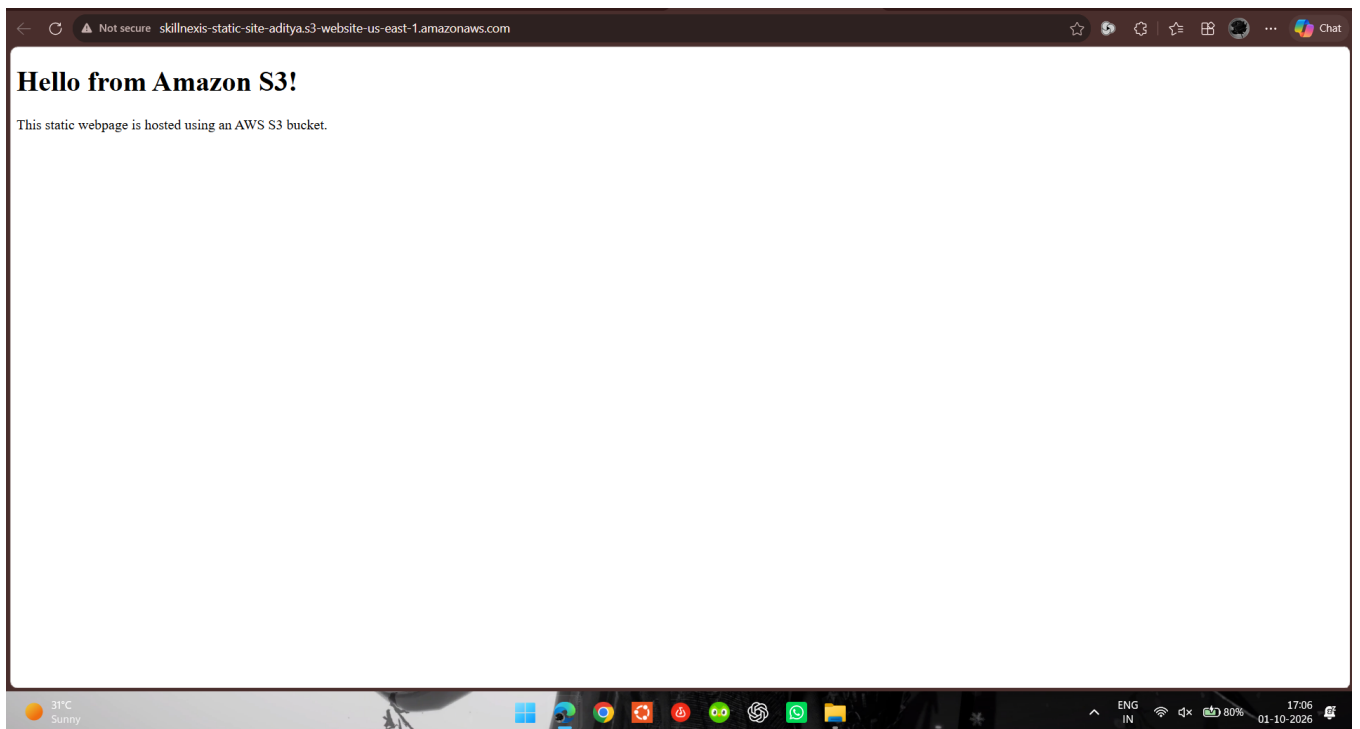
2. Static File Uploaded to S3

The screenshot displays the AWS Management Console interface for configuring a bucket policy. The left sidebar shows the navigation menu with categories like Buckets, Files, Access management and security, and Storage management and insights. The main content area is titled 'skillnexus-static-site-aditya' and features a green success message at the top: 'Successfully edited bucket policy.' Below this, there are two main sections: 'Block public access (bucket settings)' and 'Bucket policy'. The 'Block public access' section shows a toggle switch for 'Block all public access' set to 'Off'. The 'Bucket policy' section displays a JSON policy that grants 's3:GetObject' permission to the public. The policy is as follows:

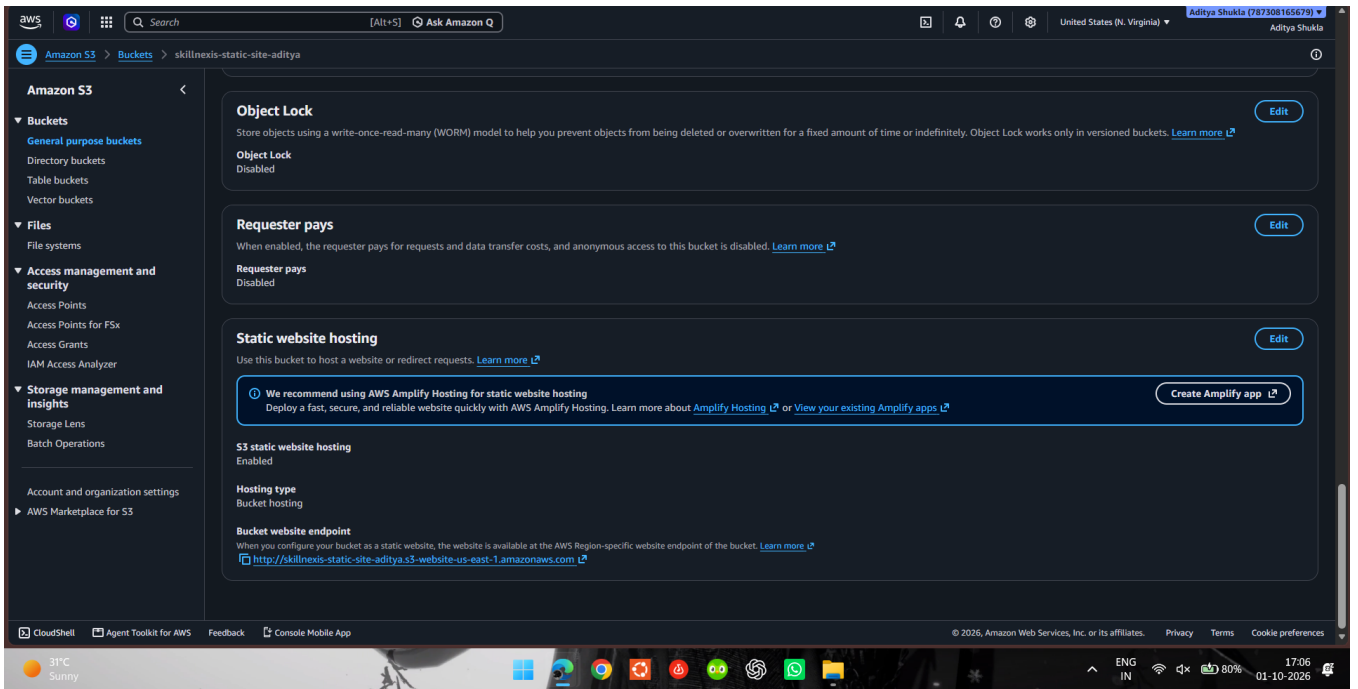
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::skillnexus-static-site-aditya/*"
    }
  ]
}
```

The bottom of the console shows the URL 'https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1' and the system status bar with weather, time, and battery information.

3. Bucket Policy Configured for Public Read Access



4. S3 Static Website Hosting Enabled



5. Hosted Static Website Successfully Running

Learning Outcome

This project demonstrates practical exposure to AWS cloud fundamentals, IAM user and permission management, Amazon S3 object storage, bucket policies, public access configuration, and deployment of a basic static website using an S3 website endpoint.

Project Completion Checklist

Task	Status
Create IAM user with S3 permissions	Completed
Create/use S3 bucket	Completed
Upload static HTML file	Completed
Configure public access and bucket policy	Completed
Enable S3 static website hosting	Completed
Verify website through S3 endpoint	Completed

Note: Public access was enabled specifically for this static website hosting exercise. For production workloads, AWS recommends avoiding unnecessary public access and following least-privilege security practices.